

Enzymes Summary Table

After a Meal

Enzyme	Function	Location in the cell	Pathway involved	Effect of Insulin	Effect of Glucagon	Allosteric Activators	Allosteric Inhibitors
Hexokinase	Phosphorylate glucose And activates it	Cytosol (Tissues)	Glycolysis	Activate Hexokinase,	Inhibits Hexokinase		G-6-P
Glucokinase	Phosphorylate glucose And activates it	Cytosol (Liver , Pancreas, Brain, RBC)	Glycolysis	Activates Glucokinase by upregulation	Inhibits Glucokinase		
PFK-2 (PFK2 – FBP2)	Phosphorylate F6P to F2,6 bisP	Cytosol	Glycolysis	Activates PFK2 kinase Increase F2,6bisP	Activates FBP2 phosphatase Decrease F2,6bisP	AMP F-2,6-BisP	ATP
Pyruvate Kinase	Generate ATP from PEP to Pyruvate	Cytosol	Glycolysis	Upregulate, de-Phosphorylate and activates	Down regulates phosphorylate and inhibits	AMP F-1,6-Bis-P	ATP Acetyl CoA Alanine
Glucose-6-P DH	From Glucose-6-P Production of NADPH	Cytosol on many tissues specially RBC and Adipose tissues and liver	HMP	Upregulates De-Phosphorylate and activates	Down-regulates Phosphorylate and inhibits		
Glycogen Synthase	Add UDP glucose to glycogenin and cleave UDP and elongate the chain	Cytosol of liver and muscles	Glycogenesis	De-Phosphorylates and activates	Phosphorylates and inhibits	ATP G-6-P Glucose	AMP (Muscle) Ca2+
Lipoprotein Lipase	Cleave TAG circulating in chylomicrons and VLDL to Glycerol and Free FA	LPL is attached to the luminal surface of endothelial cells in capillaries	After a meal in the fat Cells and some tissues like Heart / skeletal muscles and Fat cells	Upregulates also, De-Phosphorylate and activates	Down-regulates also Phosphorylate and inhibits	Activated by Apo CII	
Citrate Lyase	Cleaves cytosolic Citrate to Acetyl CoA and OAA	In the cytosol	Fatty Acid Synthesis and Cholesterol Synthesis	De-Phosphorylate and activates	Phosphorylate and inhibits		
Acetyl-CoA Carboxylase	Carboxylation of Acetyl CoA to Malonyl CoA	Cytosol of liver	Fatty Acid de-novo Synthesis	Activates by: Upregulation De-Phosphorylation	Inhibits by: Phosphorylation Down-regulation	ATP Cytosolic Citrate	AMP Palmitoyl- CoA
HMG-CoA reductase	Make Mevalonate from HMG CoA in the cytosol	In the cytosol bound to the ER	Cholesterol synthesis	Upregulates De-Phosphorylate and activates	Down-regulates Phosphorylate and inhibits	ATP Cytosolic Citrate	AMP Cholesterol
Protein Synthesis	From αKeto acid → Amino Acids → Protein	Cytosol of most tissues, particularly Muscles	Protein Synthesis (Glutamate DH and Glutamine Synthetase)	Activates	Inhibits	AMP	ATP

During Fasting

Enzyme	Function	Location in the cell	Pathway involved	Effect of Insulin	Effect of Glucagon	Allosteric Activators	Allosteric Inhibitors
Pyruvate Carboxylase	Generates OAA from Pyruvate	liver Mitochondrial	Gluconeogenesis	De-Phosphorylates and inhibits	Phosphorylates and activates	Acetyl CoA	
Fructose 1,6 bis phosphatase	De-phosphorylates F1,6bisP to F6P	liver Cytosol	Gluconeogenesis	De-Phosphorylates and inhibits	Phosphorylates and activates	ATP	ADP F-2,6-bisP
Glucose-6-Phosphatase	Generate free Glucose from Glucose-6-P	Liver ER	Gluconeogenesis Glycogenolysis	De-Phosphorylates and inhibits	Phosphorylates and activates	G-6-P	
Glycogen Phosphorylase	Phosphorylate Glycogen using inorganic phosphate and Generates G1P	Cytosol of liver and muscles	Glycogenolysis	De-Phosphorylates and inhibits	Phosphorylates and activates	AMP (Muscle) Ca ²⁺	ATP G-6-P Glucose
Hormone sensitive lipase	Hydrolyze TAG stored in the Adipose tissues (fat cells)	Intracellular enzyme found in the Adipose tissues only	During fasting in the fat Cells	De-Phosphorylate and inhibits	Phosphorylate and activates		
CPT I	Transport Activated FA to the inner membrane space	Attached to the outer mitochondrial membrane	Beta oxidation	Inhibits	Activates	NAD ⁺	NADH Malonyl CoA
HMG-CoA Synthase	Make HMG CoA from condensation of 3 Acetyl CoA	Two isoenzymes in the cytosol and mitochondrial	Liver Mitoch (ketone bodies synthesis)	Inhibits	Activates the mitochondria copy for Ketone bodies synthesis		
HMG-CoA Lyase	Cleave HMG CoA to acetoacetyl CoA and acetyl CoA	In the Mitoch of the Liver	Ketone bodies synthesis	Inhibits	Activates for Ketone Bodies synthesis		
Protein degradation and Amino Acid degradation to Alpha-Keto Acids (occurs after protein rich meal or during prolong starvation)	Diamination of Amino Acids will release free amino group + αKeto acid → Urea production	From protein to Amino Acids → TCA cycle intermediates which are used for → Glucose Synthesis or energy production	Protein Degradation Cytosol of most tissues, particularly Muscles (Glutamate DH and Glutaminase)	Inhibits	Activates (Glucagon will be high in fast state and after a protein rich meal)	AMP	ATP

Blue: Active After a meal.
Red: Active during fasting